

1. Given a signed 32-bit integer x , return x with its digits reversed. If reversing x causes the value to go outside the signed 32-bit integer range: $[-2^{31}, 2^{31}-1]$ then return 0.

Example 1

Input:

$x = 123$

Output:

321

Example 2

Input:

$x = -123$

Output:

-321

Example 3

Input:

$x = 120$

Output:

21

Example 4

Input:

$x = 1534236469$

Output:

0

2. Given an integer array `nums`, return all the triplets `[nums[i], nums[j], nums[k]]` such that $i \neq j$, $i \neq k$, and $j \neq k$, and $nums[i] + nums[j] + nums[k] == 0$. Notice that the solution set must not contain duplicate triplets.

Example 1:

Input: `nums = [-1,0,1,2,-1,-4]`

Output: `[[-1,-1,2],[-1,0,1]]`

Explanation:

$nums[0] + nums[1] + nums[2] = (-1) + 0 + 1 = 0$.

$nums[1] + nums[2] + nums[4] = 0 + 1 + (-1) = 0$.

$nums[0] + nums[3] + nums[4] = (-1) + 2 + (-1) = 0$.

The distinct triplets are `[-1,0,1]` and `[-1,-1,2]`.

Notice that the order of the output and the order of the triplets does not matter.

Example 2:

Input: `nums = [0,1,1]`

Output: `[]`

Explanation: The only possible triplet does not sum up to 0.

3. Write a function to find the longest common prefix string amongst an array of strings. If there is no common prefix, return an empty string `""`.

Example 1:

Input: `strs = ["flower","flow","flight"]`

Output: `"fl"`

Example 2:

Input: `strs = ["dog","racecar","car"]`

Output: ""

4. Given a string `s` consisting of words and spaces, return the length of the last word in the string. A word is a maximal substring consisting of non-space characters only.

Example 1:

Input: `s = "Hello World"`

Output: 5

Example 2:

Input: `s = " fly me to the moon "`

Output: 4

Example 3:

Input: `s = "luffy is still joyboy"`

Output: 6

5. Given two non-negative integers `num1` and `num2` represented as strings, return the sum of `num1` and `num2`, also represented as a string.

Note: You must not use any built-in BigInteger library or convert the inputs to integer directly.

Example 1:

Input: `num1 = "2", num2 = "3"`

Output: "5"

Example 2:

Input: `num1 = "123", num2 = "456"`

Output: "579"